

Stress Analysis

Report_Traffic_Light_without_Motor_ABS



Analyzed File:	TLA_without_motor.ipt
Autodesk Inventor Version:	2027 (Build 310192000, 192)
Creation Date:	2026-08-14, 오후 12:50
Study Author:	Kwon_Doyeong
Summary:	

Static Analysis:1

General objective and settings:

Design Objective	Single Point
Study Type	Static Analysis
Last Modification Date	2026-08-14, 오후 12:45
Model State	[Primary]
Detect and Eliminate Rigid Body Modes	No

iProperties

Summary

Title	
Subject	
Author	Kwon

Project

Part Number	TLA_without_motor
Designer	Kwon

Physical

Material	Generic
Density	1 g/cm ³
Mass	0.113991 kg
Area	43823.1 mm ²
Volume	113991 mm ³
Center of Gravity	x=-17.0173 mm y=31.6769 mm z=5.42616 mm

Note: Physical values could be different from Physical values used by FEA reported below.

Mesh settings:

Avg. Element Size (fraction of model diameter)	0.1
Min. Element Size (fraction of avg. size)	0.2
Grading Factor	1.5
Max. Turn Angle	60 deg
Create Curved Mesh Elements	Yes

Material(s)

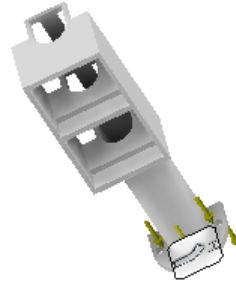
Name	ABS Plastic	
General	Mass Density	1.06 g/cm ³
	Yield Strength	20 MPa
	Ultimate Tensile Strength	29.6 MPa
Stress	Young's Modulus	2.24 GPa
	Poisson's Ratio	0.38 ul
	Shear Modulus	0.811594 GPa
Part Name(s)	TLA_without_motor.ipt	

Operating conditions

Gravity

Load Type	Gravity
Magnitude	9810.000 mm/s ²
Vector X	-0.000 mm/s ²
Vector Y	-9810.000 mm/s ²
Vector Z	-0.000 mm/s ²

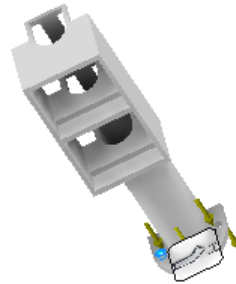
Selected Face(s)



Bearing Load:1

Load Type	Bearing Load
Magnitude	8.000 N
Vector X	0.000 N
Vector Y	-8.000 N
Vector Z	0.000 N

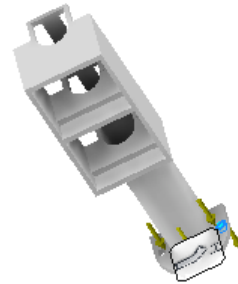
Selected Face(s)



Bearing Load:2

Load Type	Bearing Load
Magnitude	8.000 N
Vector X	0.000 N
Vector Y	-8.000 N
Vector Z	0.000 N

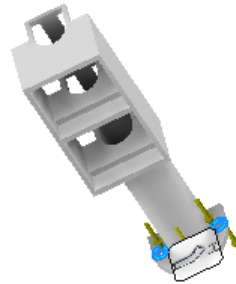
Selected Face(s)



Force:1

Load Type	Force
Magnitude	1000.000 N
Vector X	-0.000 N
Vector Y	-1000.000 N
Vector Z	-0.000 N

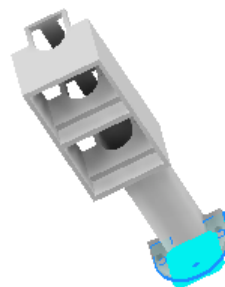
Selected Face(s)



Fixed Constraint:1

Constraint Type	Fixed Constraint
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Selected Face(s)



Results

Reaction Force and Moment on Constraints

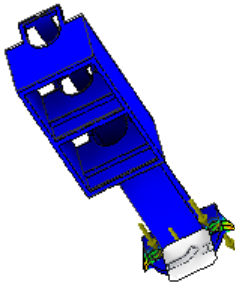
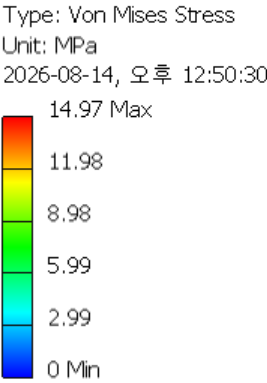
Constraint Name	Reaction Force		Reaction Moment	
	Magnitude	Component (X,Y,Z)	Magnitude	Component (X,Y,Z)
Fixed Constraint:1	1016.31 N	0 N	3.13222 N m	3.13207 N m
		1016.31 N		0 N m
		0 N		-0.0307345 N m

Result Summary

Name	Minimum	Maximum
Volume	29936.9 mm^3	
Mass	0.0317331 kg	
Von Mises Stress	0.0000275011 MPa	14.9719 MPa
1st Principal Stress	-9.88615 MPa	3.72654 MPa
3rd Principal Stress	-21.102 MPa	0.517566 MPa
Displacement	0 mm	0.0196008 mm
Safety Factor	1.33583 ul	15 ul

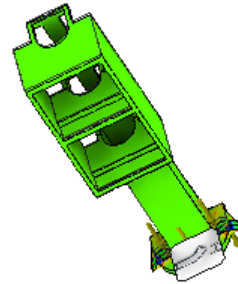
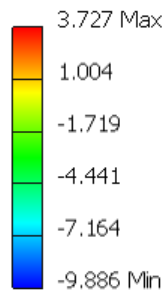
Figures

Von Mises Stress



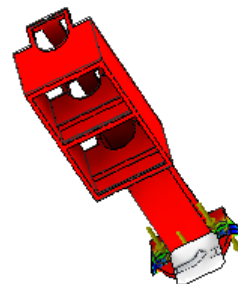
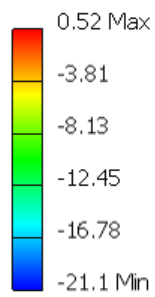
1st Principal Stress

Type: 1st Principal Stress
Unit: MPa
2026-08-14, 오후 12:50:31



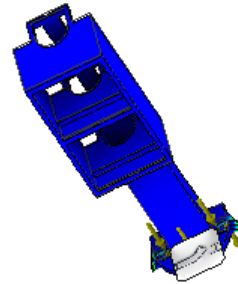
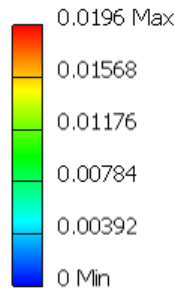
3rd Principal Stress

Type: 3rd Principal Stress
Unit: MPa
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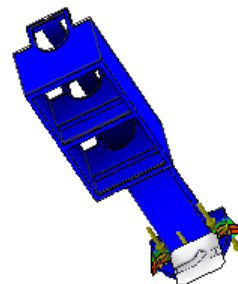
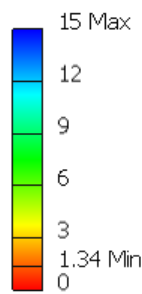
Displacement

Type: Displacement
Unit: mm
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Safety Factor

Type: Safety Factor
Unit: ul
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